

Concept of Geo-informatics

- Use of computer has become cheaper in last 15 years.
- Reduction in the manufacturing cost of hard ware.
- These changes in hardware technology have gone in parallel with better data conversion, better software for image manipulation and interpretation and improvement in database management system.
- The innovation and development in computer, communication and software has contributed towards the growth of information technology.

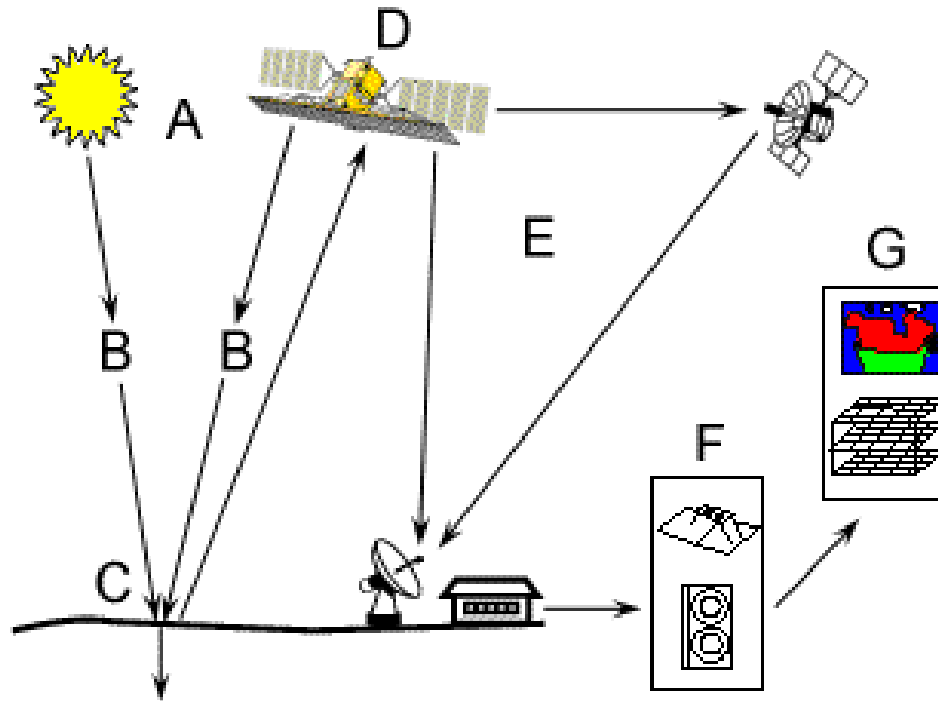
- The net result of these changes is that it is now relatively easy to create, store, retrieve, and analyze large quantities of spatial and non-spatial data for specific purpose.
- Using Remote Sensing (RS), Global Positioning System (GPS) and Geographical Information System (GIS), one tries to gather information to take a decision as required by the management/administration which leads to concept of Geo-informatics.

Remote Sensing (RS)

Remote sensing is the science (and to some extent, art) of acquiring information about the earth surface without actually being in contact with it.

This is done by sensing and recording the reflected or emitted energy and processing, analyzing, and applying that information.

Remote Sensing Process Components



Energy Source or Illumination (A)

Radiation and the Atmosphere (B)

Interaction with the Target (C)

Recording of Energy by the Sensor (D)

Transmission, Reception, and Processing (E)

Interpretation and Analysis (F)

Application (G)

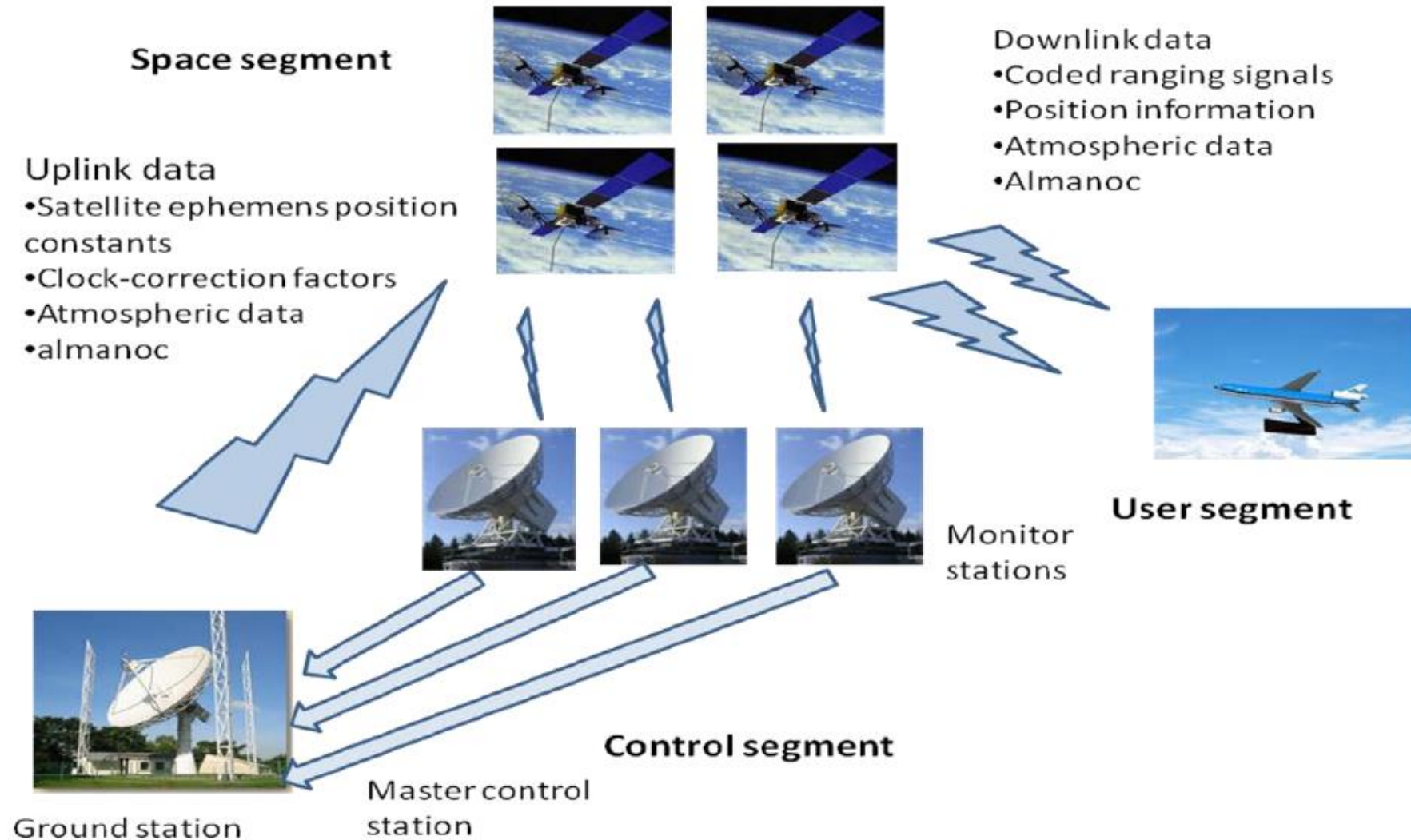
Global positioning system (GPS)

Global positioning system is an earth-orbiting Satellite based system that provides signals anywhere on or above earth, 24 hours a day, round the year, and irrespective of weather

It can be used to determine precise time and the position of a GPS receiver in three dimensions.

This technology is increasingly used as input for GIS particularly for precise positioning of geo-spatial data and for collection of data from the field.

GPS segments



Geographical Information System (GIS)

A form of Information Technology (IT) is Geographic Information System (GIS).

It is described as: “An organized collection of computer hardware, software, geographic data and personnel designed to efficiently capture, store, update, manipulate, analyze, and display all forms of geographically referenced information”.

Components of GIS

